INCH-POUND
A-A-50728B (USAF)
19 March 2021
SUPERSEDING
A-A-50728A
21 April 2010

COMMERCIAL ITEM DESCRIPTION

FREQUENCY SELECTIVE LEVELMETER AND SPECTRUM DISPLAY

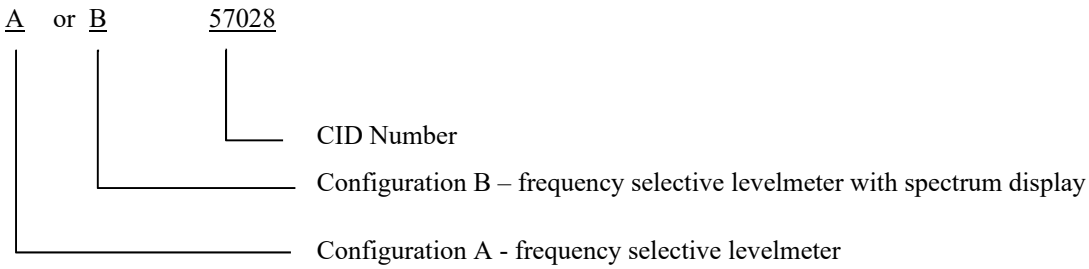
The General Services Administration has authorized the use of this commercial item description by all Federal agencies.

1. SCOPE. This commercial item description (CID) covers general requirements for a frequency selective levelmeter and frequency selective levelmeter with spectrum display test equipment or equipment, in either configuration. Only those manufacturers that meet or surpass the minimum requirements, as specified herein, are qualified per this CID. The frequency selective levelmeter and/or frequency selective levelmeter and spectrum display (FSLSD) is also referenced as the equipment, or the test equipment.

2. CLASSIFICATION.

2.1 Classification. The equipment covered by this CID shall be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800.

2.2 Part Numbering. The configuration will be referenced as follows:



3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications as specified herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Area B, Wright Patterson AFB, OH 45433-7017 or emailed to Engineering.Standards@wpafb.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3.1 Safety. The equipment shall comply with all electrical safety requirements as specified in the Underwriters Laboratories (UL) Safety Standard for Surge Protection Devices UL 1449. The equipment shall comply with the safety requirements as specified in MIL-PRF-28800.

3.1.1 Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliampere root mean square (mA-rms), alternating current (AC) and direct current (DC), in any accessible conductive parts of equipment. This includes control shafts with knobs broken or removed and recessed calibration or adjustment controls to either pole of the power source for any position of the power switch(es).

3.1.2 Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated at least four times the operation voltage.

3.1.3 Ventilation openings.

3.1.3.1 Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

3.1.3.2 Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

3.2 Parts, materials, and processes. Except for marking, the FSLSD shall comply with the parts, materials, and process requirements of MIL-PRF-28800. Nonstandard parts approval requests shall be in accordance with MIL-PRF-28800. Front panel marking shall be in accordance with MIL-PRF-28800.

3.2.1 Restricted material. The equipment shall comply with the restricted material requirements of MIL-PRF-28800, including the requirement for mercury and radioactive materials.

3.2.2 Combination of material. This equipment shall contain no combination of materials, which may cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.3 Design and construction. Unless otherwise specified herein, the equipment shall comply with design and construction requirements as specified in MIL-PRF-28800**.** The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits, where construction is susceptible to leakage or corrosion. Indicators and associated controls shall be in close proximity to each other. All potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

3.3.1 Type of construction. Construction of the FSLSD shall be of a monolithic or a separate component design.

3.3.2 Modular design restrictions. The modular design concept for the FSLSD is acceptable with the following restrictions:

3.3.2.1 The FLSD shall consist of a single unit with one or more separate modules.

3.3.2.2 Each module shall contain a power circuit protection fuse.

3.3.2.3 Each module plug-in card shall be keyed and mated with the back plane receptacle, so that modules may not be randomly placed into the mainframe.

3.3.2.4 Modules shall not be removable by access through the front or rear of the mainframe with the unaided hand.

3.3.2.5 The FSLSD shall come with one blank space cover and one extender board.

3.3.2.6 A package part number shall be assigned to the FSLSD as a whole system. This system part number shall be marked on the mainframe identification plate and all plug-in(s).

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3.3.3 Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female.

3.3.4 Internal cooling. The equipment shall be capable of operating continuously for twenty four hours at 55°C. If forced air is used, it shall be as specified in MIL-PRF-28800.

3.3.4.1 Forced air-cooling motor. DC brushless motors shall be utilized for the forced air-cooling motor inside the equipment.

3.3.4.2 Air filter. Equipment utilizing a cooling scheme, which exhausts air from the equipment through the cooling fan, shall not require an externally accessible air filter.

3.3.4.3 Thermal switch. A thermal switch shall be provided, which will interrupt circuit power to the FSLSD, when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10 ºC hysteresis drop from this tripping temperature must occur before circuit power is again applied.

3.4 Electrical power sources. The equipment shall operate from nominal commercial, military and shipboard power sources with a nominal voltage of 120 volts root mean square (Vrms) single-phase AC (±10%) at line frequencies of 50 hertz (Hz) and 60Hz in accordance with MIL-PRF-28800.

3.4.1 Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single-phase input.

3.4.2 Power consumption. Maximum power consumption of all components combined shall not exceed 150 watts.

3.4.3 Fuse(s). Fuse(s) shall be provided on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

3.4.4 Electrical surge, noise and transient voltage protection. The FSLSD shall be able to reject power line electrical noises in the 30 Hz to 10 kilohertz (kHz) region as specified in MIL-STD-461 with noise levels below the 70 dB. The FSLSD shall be able to provide surge protection that meets the requirements as specified in UL 1449. The FSLDS shall be able to meet the testing and measurement requirements as specified in the International Electrotechnical Commission (IEC) EN 61000-4-5. Transient state conditions shall meet the classification requirements as specified in MIL- PRF- 28800.

3.5 Dimensions and weight. The size and weight of individual components shall be consistent with current commercial capabilities.

3.5.1 Equipment dimensions. The nominal dimensions per component of the FSLSD should be approximately 7 inches high x 17 inches wide x 18 inches deep.

3.5.2 Equipment weight. The weight including all accessories, excluding manual, per component shall not exceed 50 pounds (22.68 kilograms).

3.6 Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the enclosure requirements as specified in MIL-PRF-28800.

3.6.1 Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

3.6.2 Tilt bail handle. A tilt bail handle shall be provided in lieu of the hinged metal handles in accordance with MIL-PRF-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

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3.6.3 Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front cover storage.

3.7 Marking and identification.

3.7.1 Marking. The visible surface adjacent to panel facilities, such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation indicating the purpose or use of the part.

3.7.2 Marking identification. Marking identification process requirements may be used as specified in MIL- PRF-28800. Providing the marking and identification processes are capable of passing the front panel marking test, when tested in accordance with this CID.

3.7.3 Items to be marked, if applicable.

3.7.3.1 Volume and weight, including accessories, shall be marked on three external surfaces of the enclosure.

3.7.3.2 The type designation of the Army/Navy (AN) nomenclature of the FSLSD shall be stenciled on the exterior or main frame on at least three adjacent surfaces or on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

3.8 Environmental requirements.

3.8.1 Temperature and humidity requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements as specified in MIL-PRF-28800.

3.8.1.1 Storage temperature range. The equipment shall meet and/or exceed manufacturers' specifications for stored equipment after temperature and humidity tests have been performed for instruments in the class 3 range as specified in MIL-PRF-28800.

3.8.1.2 Operating temperature range. The equipment shall be able to meet the specifications listed herein and perform within the operating temperature range of 32 to 122°F (0 to 50°C), after temperature and humidity tests have been performed for the Class 3 equipment classification as specified in MIL-PRF-28800.

3.8.2 Altitude requirements. The equipment shall meet its operating performance and accuracy requirements while at an altitude of 15,090 feet (4,600 meters). The equipment shall meet its performance and accuracy requirements for operating and non-operating storage at this altitude as specified in MIL-PRF-28800.

3.8.3 Vibration requirements. The equipment shall meet the Class 3 vibration requirements as specified in MIL- PRF- 28800.

3.8.4 Mechanical shock requirements.

3.8.4.1 Functional shock test. The equipment shall meet the shock testing and pulse requirement as specified in MIL- PRF-28800.

3.8.4.2 Transit drop. The equipment shall meet transit drop requirement as specified in MIL-PRF-28800.

3.8.4.3 Bench handling. The equipment shall meet the specified performance and accuracy requirements as specified in MIL-PRF-28800.

3.8.4.4 Shock, high impact. The equipment shall meet the high impact shock test requirements as specified in MIL-PRF-28800.

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3.8.5 Electromagnetic interference (EMI). The FSLSD EMI emissions shall be in accordance with the radiated emission and susceptibility requirements as specified in MIL STD 461 for radiated emissions (RE) RE102 and radiated susceptibility (RS) RS103, IEC 61000-4 thru IEC-61000-6, and the radiation absorption and radiation limits requirements as specified in the Federal Communications Commission (FCC) Title 47 of the Code of Federal Regulations (14 CFR), section 15.109, Radiated emission limits and section 15.209, Radiated emission limits; general requirements.

3.8.6 Magnetic environment. The equipment shall comply with the magnetic environment requirements as specified in MIL-PRF-28800.

3.9 Reliability. The design of the FSLSD shall be such that under normal use and operation the mean time before failure (MTBF) exceeds 1,500 hours of operation with a statistical certainty equal or greater than of the FSLSD shall be 95% in accordance with MIL-PRF-28800. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL- PRF-28800. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding the requirements as specified in MIL- PRF-28800.

3.9.1 Mean time between failures. The equipment MTBF shall be 2000 hours. The lower test MTBF shall be 2000 hours and the upper test MTBF shall be 4000 hours.

3.9.2 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

3.9.2.1 Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, 104°F (40ºC) for a minimum of 100 hours. Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported, but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be conducted at selected times during the burn-in test.

3.9.2.2 Temperature test procedure. The temperature test procedure shall expose the equipment to five independent tests as specified in MIL-PRF-28800 for class 3 equipment.

3.10 Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

3.11 Maintainability. The FSLSD shall meet the human engineering factors of ergonomics for operational and maintainability criteria providing access to controls, internal components, and adjustments in accordance with MIL- STD 1472 and MIL-PRF-28800.

3.12 Interchangeability. The FSLSD shall meet the interchangeability requirements as specified in MIL- PRF- 28800.

3.13 Digital interface. The equipment shall meet the digital interface requirements as specified in MIL- PRF- 28800.

3.14 Performance characteristics. Unless otherwise specified, all Vernier controls shall be in the calibrated position. All performance requirements shall apply after a 30-minute warm up period in a laboratory ambient operating temperature.

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3.14.1 General requirements.

3.14.1.1 Configuration A 50728. Configuration A system shall consist of a frequency selective levelmeter that meets all performance characteristics.

3.14.1.2 Configuration B 50728. Configuration B system shall consist of an FSLSD that meets all performance characteristics.

3.14.2 Electrical and functional requirements. This section shall reflect the minimum characteristics and accuracies for the frequency selective levelmeter.

3.14.2.1 Frequency.

3.14.2.1.1 Range. Equipment shall be capable of responding to signals within a frequency range from 1 kHz to 25 megahertz (MHz) with an input impedance as follows:

75 ohms	1 kHz to 25 MHz
124 ohms	60 kHz to 11.5 MHz
135 ohms	10 kHz to 3.2 MHz
600 ohms	1 kHz to 650 MHz

3.14.2.1.2 Display. Equipment shall have the capability of allowing frequencies to be displayed linearly on the horizontal axis of a cathode ray tube display.

3.14.2.1.3 Tuning. Equipment shall be capable of digitally being set to a selected frequency. It shall have the option of being swept about this center frequency manually.

3.14.2.1.4 Manual tuning. Equipment shall have the capability of being tuned manually by both an individual frequency decade control and a single knob tuning control.

3.14.2.1.5 Decade frequency selector. The decade frequency selector shall consist of six individual decade selectors, each individually selectable from 0 to 9, and one selectable from 0 to 2 in the following range:

10	per step to 9
100	per step to 9
1	per step to 9
10	per step to 9
100	per step to 9
1	per step to 9
10	per step to 2

3.14.2.1.6 Resolution. The equipment shall be capable of resolving frequencies with an accuracy of 10 Hz.

3.14.2.1.7 Accuracy. The equipment shall have an accuracy of ±1 parts per million (ppm) ±20 Hz from 68 to 86°F (20 to 30ºC) and ±2 ppm from 32 to 113°F (0 to 45ºC).

3.14.2.1.8 Stability. The equipment shall be stable to within 5 ppm per year.

3.14.2.2 Level.

3.14.2.2.1 Range. The equipment shall have a level measuring range from -129 decibel-milliwatts (dBm) to +12 dBm over the specified frequency range referenced to 0 dBm on meter.

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3.14.2.2.2 Indicator. The equipment shall have an indicator showing what level measurement range has been selected and it shall be by means of a 3 digit light emitting diode (LED) display.

3.14.2.2.3 Resolution. The equipment shall be capable of resolving levels in 1 dBm and 10 dBm steps and shall include a continuous Vernier control with a calibrated mode.

3.14.2.2.4 Modes. The equipment shall have a noise mode and a tone mode with the following ranges:

Tone level range	-89 dBm to 10 dBm for meter zero
Noise level range	-109 dBm to -10 dBm for meter zero

3.14.2.2.5 Accuracy. Accuracy of the equipment after internal calibration at each measurement frequency shall be as follows:

75 ohms		
Tone mode	25ºC ±5ºC	
75 Hz Bandwidth		
10 KHz to 18.6 MHz		
-69 dBm to +20 dBm		±0.15 dB
-89 dBm to -69 dBm		±0.20 dB
18.6 MHz to 25 MHz		
-69 dBm to +10 dBm		±0.25 dB
-89 dBm to -69 dBm		±0.30 dB
1 KHz to 10 KHz		
-69 dBm to +10 dBm		±0.30 dB
-89 dBm to -69 dBm		±0.70 dB
Tone mode	0ºC to 45ºC	±0.10 dBm additional
Noise mode		
75 Hz Bandwidth		
10 KHz to 100 KHz		
-99 dBm to -10 dBm		±0.40 dB
-109 dBm to -99 dBm		±0.90 dB
100 KHz to 18.6 MHz		
-99 dBm to -10 dBm		±0.25 dB
-109 dBm to -99 dBm		±0.35 dB
18.6 MHz to 25MKHz		
-99 dBm to -10 dBm		±0.30 dB
-109 dBm to -99 dBm		±0.50 dB
Noise mode	0ºC to 45ºC	±0.10 dBm additional
124, 135, 600 ohms		
	25ºC to ±5ºC	±0.10 dB in addition to 75 ohms
	0ºC to ±45ºC	±0.15 dB in addition to 75 ohms

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3.14.2.3 Input.

3.14.2.3.1 Impedance. The equipment shall have four values of input impedances. They shall be 75 ohms in the unbalanced state and 124, 135, and 600 ohms in the balanced standard unit state. All four values shall be push button selectable and shall be mounted on the front panel.

3.14.2.3.2 Return loss. Return loss for the equipment shall equal or better the following:

Tone mode	75 ohms	
-79 dBm to +10 dBm		30 dB
-89 dBm to -79 dBm		2 dB
Noise mode		
-89 dBm to +10 dBm		30 dB
-99 dBm to -89 dBm		20 dB
Tone mode	124, 135, and 600 ohms	
-79 dBm to +10 dBm		30 dB except for 1st octave of frequency range 27 dB including 1st octave of frequency range
-89 dBm to -79 dBm		20 dB
Noise mode	124, 135, and 600 ohms	
-89 dBm to +10 dBm		30 dB except for 1st octave of frequency range 27 dB including 1st octave of frequency range
-99 dBm to -89 dBm		20 dB

3.14.2.3.3 Overload indicator. The equipment shall have an overload indicator light. It shall light up when high level inputs as follows are present:

For 1800 channels	
Tone Mode	25 dB overdrive typical
Noise Mode	45 dB overdrive typical

3.14.2.3.4 Signal indicator. The equipment shall have a signal indicator light that is activated when a signal equivalent to -8 dB is present.

3.14.2.4 Bandwidth. The equipment shall provide three bandwidths. Filters shall have the following selectivity:

75 Hz	75 Hz ±10%	350 Hz
1.75 KHz (equivalent noise bandwidth)	2.3 KHz ±10%	4.6 KHz
3.1 KHz	3.1 KHz ±10%	6.2 KHz

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3.14.2.5 Internal distortion.

3.14.2.5.1 Harmonic. The harmonic distortion for the equipment shall be 75 dB below the fundamental for a 60 dB sensitivity increased from 10 KHz to 25 MHz.

3.14.2.5.2 Noise power ratio (NPR). The NPR for the equipment shall be a minimum of 54 dB for a 40 dB sensitivity increase for an equivalent noise power, equivalent to 1800 channels. For an equivalent 600 channels, NPR shall be a minimum of 56 dB for a 40 dB sensitivity increase.

3.14.2.6 Intermediate frequency (IF). The equipment shall provide three IFs with an IF and image rejection of 80 dB. These three IFs shall be:

IF 1	50.7 MHz
IF 2	10.7 MHz
IF 3	80 MHz

3.14.2.7 Audio output. The equipment shall provide audio monitor jacks for external test equipment.

3.14.2.7.1 Speaker. The equipment shall provide an internal speaker connected to the audio output.

3.14.2.7.2 Control. Both the equipment's speaker and audio output monitor audio level shall be controlled by a volume adjustment knob located on the front panel.

3.14.2.7.3 Functions. The equipment shall provide upper side band (USB), lower side band (LSB) and amplitude modulation (AM) functions. These functions shall be either push button selectable or knob selectable. Control or controls shall be located on the front panel.

3.14.2.8 Residual phase jitter. The residual phase jitter for the equipment shall with a 20 dB overdrive to the meter, be 0.5º nominal for signals 20 Hz to 300 Hz from the carrier.

3.14.2.9 Tracking signal output. The equipment shall provide a tracking signal output from rear panel at 0 dBm ±0.25 dB into 75 ohms impedance from 1 KHz to 25 MHz.

3.14.2.10 Recorder output. The equipment shall provide 0.8 V DC typical into 1 K ohm impedance for 0 dB on meter for recorder operation.

3.14.2.11 Time base output. The equipment shall provide a time base output from the rear panel. Frequency shall be 10 MHz with a level of -10 dBm ±4 dB into 50 ohm impedance.

3.14.2.12 Meter. The equipment shall have a meter located on the front panel to measure the level.

3.14.2.12.1 Scale. The meter shall have both a normal and an expanded scale. Normal scale shall be from +2 dBm to -20 dBm and the expanded scale shall be from -1.0 dBm to +1.0 dBm. Minimum resolution on the expanded band shall be 0.05 dB.

3.14.2.12.2 Linearity. The meter shall have a linearity error of less than ±.02 dB in the expanded scale and less than ±0.1 dB from +2 dBm to -10 dBm and less than ±0.7 dB from -30 dBm to -20 dBm in the normal scale.

3.14.2.13 Calibration. Equipment shall have an internal calibration circuit. It shall provide its own calibration signals.

3.14.2.13.1 Level source. A precision level source shall be provided at each discrete measurement frequency available from the instrument.

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3.14.2.13.2 Control. The equipment shall provide a front panel push button and level control that shall make all the necessary connections for calibration. No external cabling shall be allowed.

3.14.2.14 Probe power. The equipment shall provide a front panel output of 15 volts direct current (VDC), 50 mA maximum for a high impedance probe.

3.14.3 Electrical and functional requirements. The following electrical and functional requirements shall reflect the minimum characteristics and accuracies acceptable for the spectrum display.

3.14.3.1 Compatibility. The equipment shall be compatible with the frequency selective levelmeter described in paragraph 2.2.

3.14.3.2 Frequency.

3.14.3.2.1 Range. The equipment shall be capable of displaying frequencies with 1 KHz to 25 MHz.

3.14.3.2.2 Sweep width. The equipment shall provide three sweep widths capable of the following:

3.14.3.2.2.1 Continuous or recurring sweep.

3.14.3.2.2.2 A single sweep that shall be activated by a push button on the front panel.

3.14.3.2.2.3 A manual sweep that shall be provided by means of a tuning control.

3.14.3.2.3 Sweep width calibration. The sweep width calibration shall be provided by the equipment using a nominal 4 KHz per channel, in channels, and in KHz/division for six horizontal sweep scales.

1
12
60
600
1800
3600

3.14.3.2.4 Sixth sweep mode. The sixth sweep mode shall sweep the entire frequency range of the instrument. The above scales represent deflection from the furthest left to right graticules of the cathode ray tube.

3.14.3.2.5 Resolution. Resolution for the equipment shall equal or better the following:

Signal level separation	0 dB	20 dB	50 dB
25	250	350	500
10	100	150	200
3	30	40	50
300	5	10	15
60	1	2	4
4	100	200	500

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3.14.3.3 Level.

3.14.3.3.1 Range. The equipment shall be capable of displaying levels in the following ranges:

Total	-109 dBm to +10 dBm (for zero level on display)
Display	60 dB logarithmic (linear dB scale)
	+40 dB to -20 dB (referred to 0 dB reading on level meter)

3.14.3.3.2 Accuracy. The equipment shall have the same accuracy displayed on the level meter in paragraph 3.14.2.2.1.

3.14.3.3.3 Frequency response. The frequency response for the equipment shall be within ±0.1 dB for full frequency span of 1 KHz to 25 MHz.

3.14.3.3.4 Display linearity. The equipment shall have a vertical linearity of ±0.5 dB in the decibels/centimeter (dB/cm) mode and ±2 dB in the 10 dB/cm mode.

3.14.3.4 IF bandwidths. The equipment shall have the following sweep widths.

at 3		
25	10 MHz and 3 MHz	5
300		1.5
60		400
4		45.0

3.14.3.5 Sweep rate. The equipment shall be capable of providing the following sweep rates:

Sweep widths	Sweep rate
25	0.9 Hz (2500 MS period)
10	1.0 Hz (1000 MS period)
3	
300	2.2 Hz (450 MS period)
60	
4	0.5 Hz (2000 MS period)

3.14.3.6 High resolution, multi-touch display. It shall display level versus frequency.

3.14.3.7 High level alarm output. The equipment shall provide a high level alarm output. It shall be by means of a set of dry contacts on the rear panel rated at 1 amp at 125 VAC.

3.14.3.8 High level alarm trigger level. The high level alarm shall trigger at the presence of any of the following:

Sweep widths	4 KHz to 10 MHz
10 dB/div	+1.0 dB graticule ±2.5 dB
1 dB/div	+0.1 dB graticule ±1 dB

3.15 Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

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4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered, recycled, environmentally preferable, or biobased content to the maximum extent practicable in accordance with Federal Acquisition Regulation (FAR) paragraphs 23.403 and 23.404(b).

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR Part 23.404(b) "Exemptions" applies and states the (1) Agency affirmative procurement programs must require 100 percent of purchases of EPA or USDA-designated items contain recovered material or biobased content, respectively, unless the item cannot be acquired: (i) Competitively within a reasonable timeframe; (ii) Meeting reasonable performance standards; or (iii) Subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability may compete by submission of bid samples. The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Military specifications, standards, and handbooks. Military specifications, standards, and handbooks referenced herein may be obtained at https://assist.dla.mil.

7.1.2 FAR and CFR. Electronic copies of the FAR may be obtained from https://www.acquisition.gov. Electronic copies of the CFR are available at http://www.ecfr.gov/cgi-bin/ECFR?page=browse.)

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7.1.3 IEC standards. IEC standard products may be obtained at http://iec.ch/.

7.1.4 UL standards. UL standard products may be obtained at https://ulstandards.ul.com/access-standards/.

MILITARY INTERESTS:

Custodian:
Air Force – 84

Preparing Activity:
Air Force - 84

Agent:
Air Force - 110

Project 6625-2021-002

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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